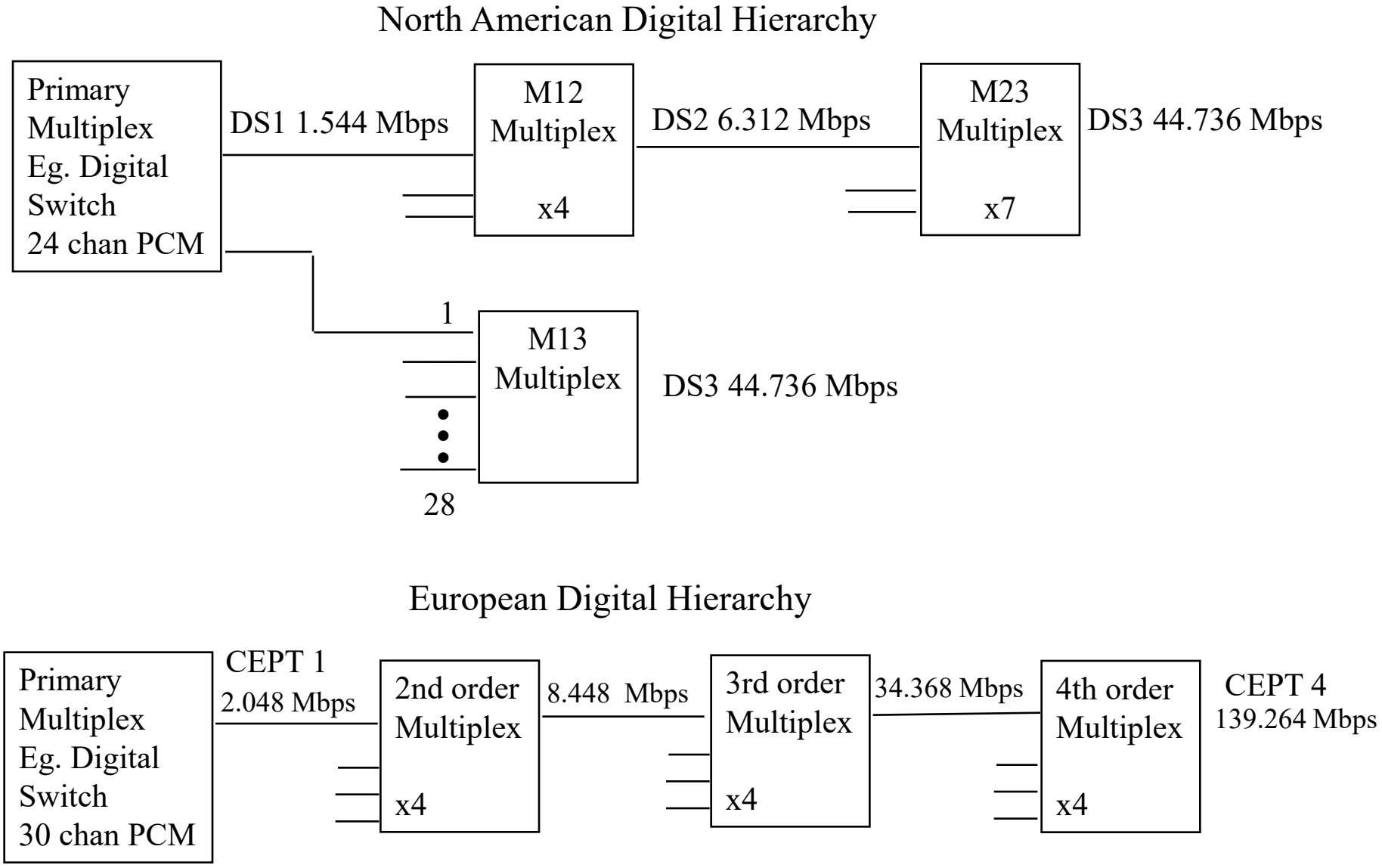
Telecommunication Systems
ICE 3109

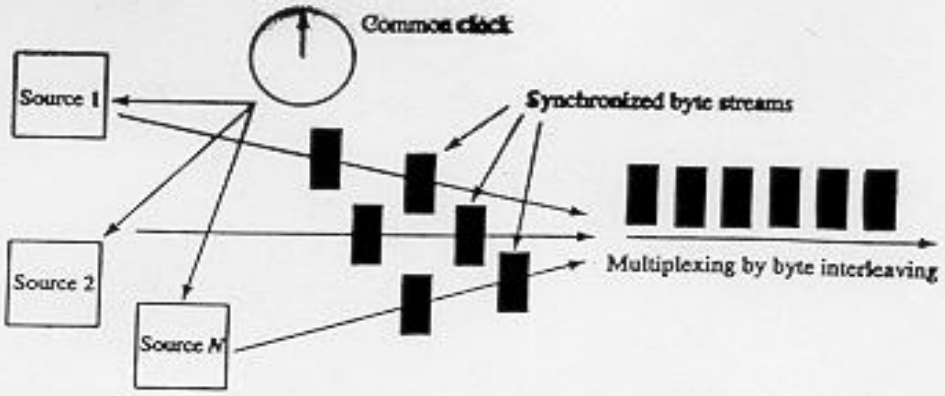
Lecture 12
Synchronous Optical Network
SONET

Telephone Networks {Brief History}

- Digital carrier systems
  - The hierarchy of digital signals that the telephone network uses.
  - Trunks and access links organized in **DS (digital signal) hierarchy**
  - Problem: rates are not multiples of each other.
- In the 1980's Bellcore developed the Synchronous Optical Network (SONET) standard.
- Previous efforts include: **ISDN** and **BISDN.**

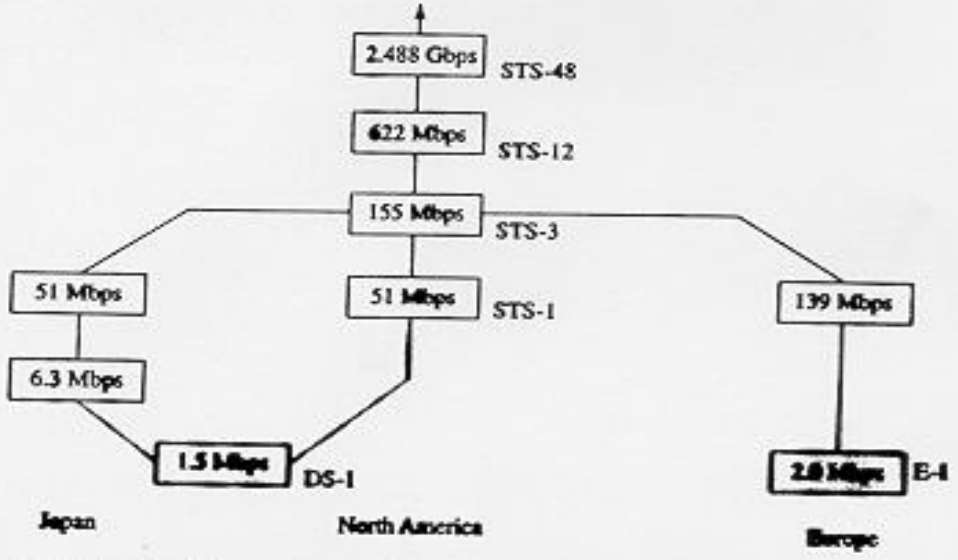


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4.2 FIGURE

SONET sources are synchronized to a common master clock. Different streams are multiplexed by byte interleaving.



4.3 FIGURE

The STS-n signal has a rate equal to $n \times 51.84$ Mbps. In Europe the hierarchy starts at 155.52 Mbps. All the standards become compatible at speeds of 155 Mbps.

Medium	Signal	Voice circuits	North America	Japan	Europe
T-1 paired cable	DS-1	24	1.5	1.5	2.0
T-1C paired cable	DS-1C	48	3.1		
T-2 paired cable	DS-2	96	6.3	6.3	8.4
T-3 coax, radio, fiber	DS-3	672	45.0	34.0	32.0
Coax, waveguide, radio, fiber	DS-4	4032	274.0		

1.2 TABLE

Digital carrier systems. This is the hierarchy of digital signals that the telephone network uses. Note that the bit rate of a DS-1 signal is greater than 24 times the rate of a voice signal (64 Kbps) because of the additional framing bits required.

Carrier	Signal	Rate in Mbps
OC-1	STS-1	51.840
OC-3	STS-3	155.520
OC-9	STS-9	466.560
OC-12	STS-12	622.080
OC-18	STS-18	933.120
OC-24	STS-24	1244.160
OC-36	STS-36	1866.240
OC-48	STS-48	2488.320

1.3 TABLE

SONET rates. The rates of multiplexed STS-1 signals are exact multiples; no additional framing bits are used.

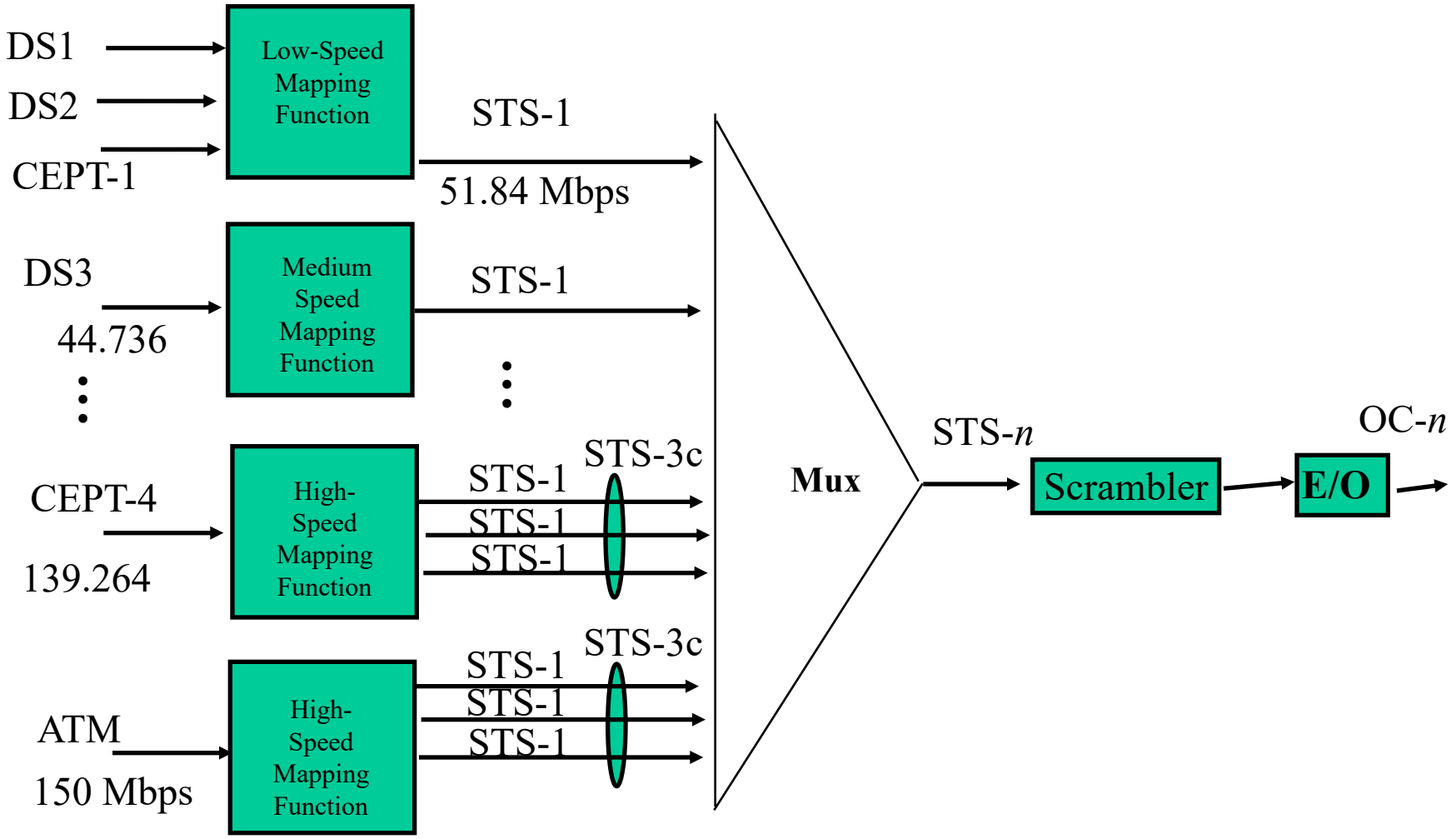
SONET

- Transmission links of the telephone network have been changing to *SONET* where rates are arranged in **STS** (Synchronous Transfer Signal) hierarchy.
- The hierarchy is called **SDH** (Synchronous Digital Hierarchy) defined by CCITT.
- It is an ITU standard.

SONET

SONET:: encodes bit streams into optical signals propagated over optical fiber. SONET defines a technology for carrying many signals of different capacities through a synchronous, flexible, optical hierarchy.

- A bit-way implementation providing end-to-end transport of bit streams.
- All clocks in the network are locked to a common master clock so that simple TDM can be used.
- Multiplexing done by *byte interleaving*.
- *SONET* is backward compatible to DS-1 and E-1 and forward compatible to ATM cells.
- Demultiplexing is easy.

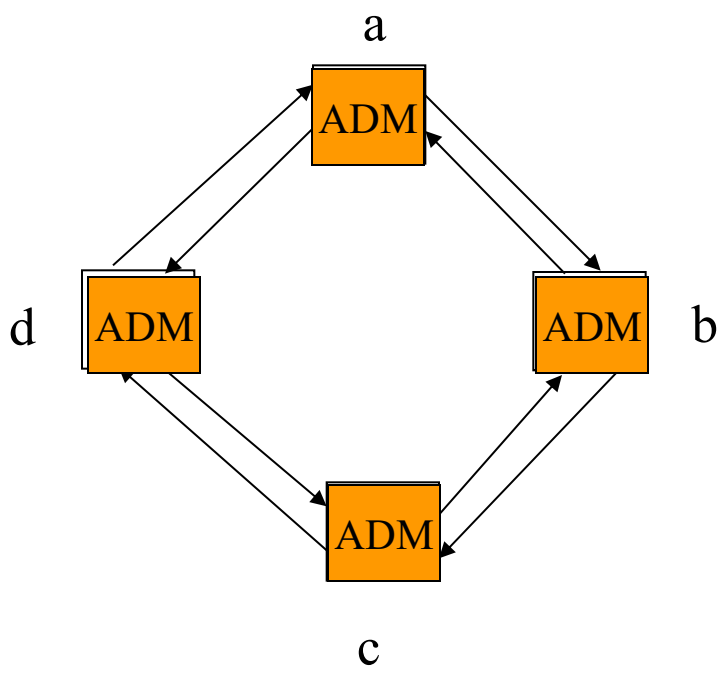


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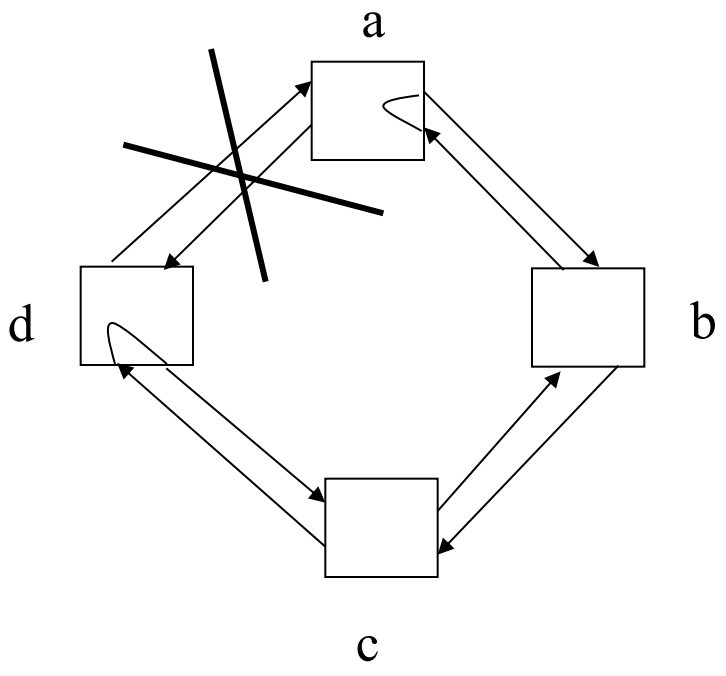
Figure 4.8

SONET Architecture

- SONET topology can be a mesh, but most often it is a dual ring.
- Standard component of SONET ring is an **ADM** (Add/Drop Multiplexer)
  - Drop one incoming multiplexed stream and replace it with another stream.
  - Used to make up bi-directional line switching rings.

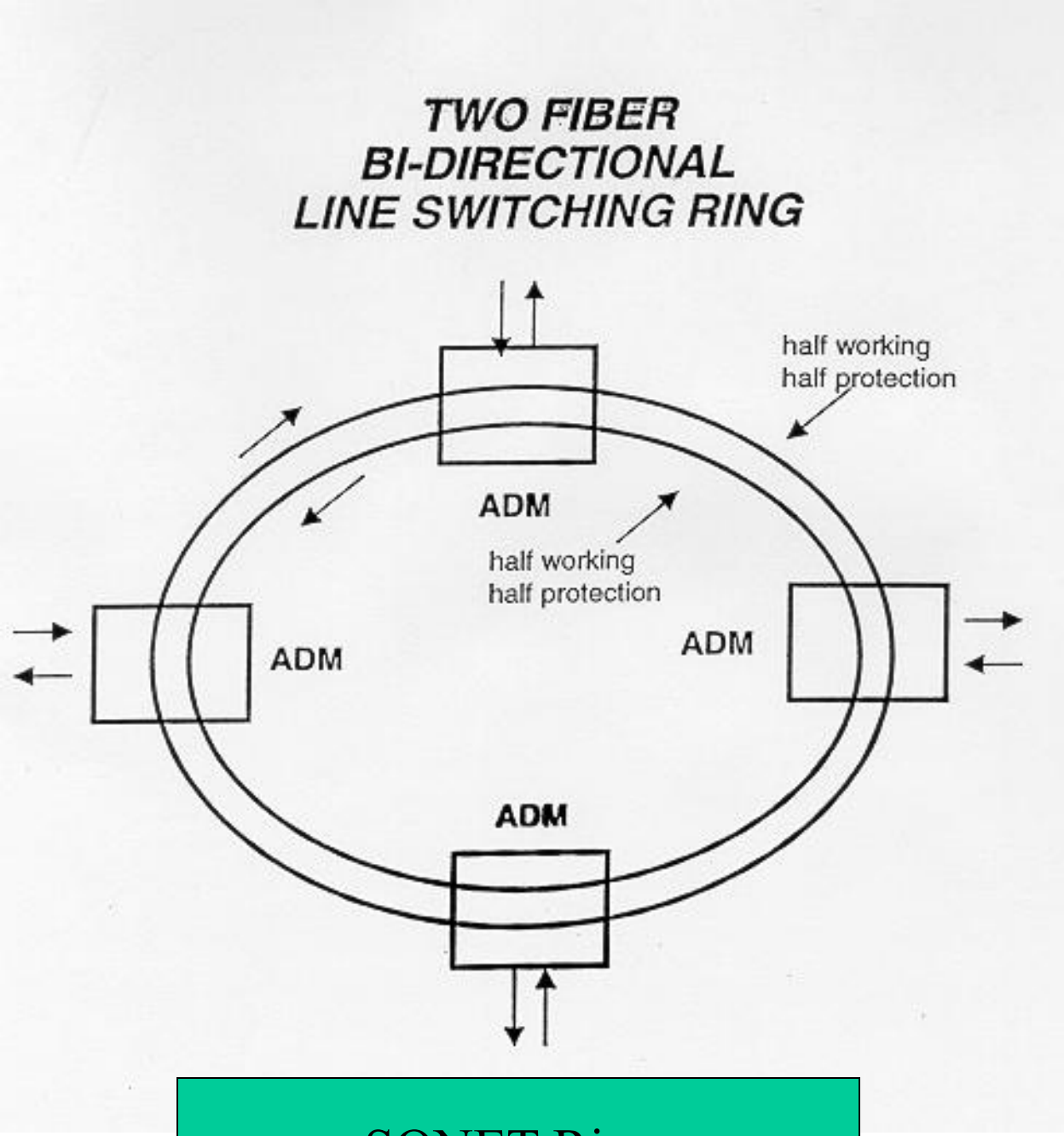


(a) Dual ring



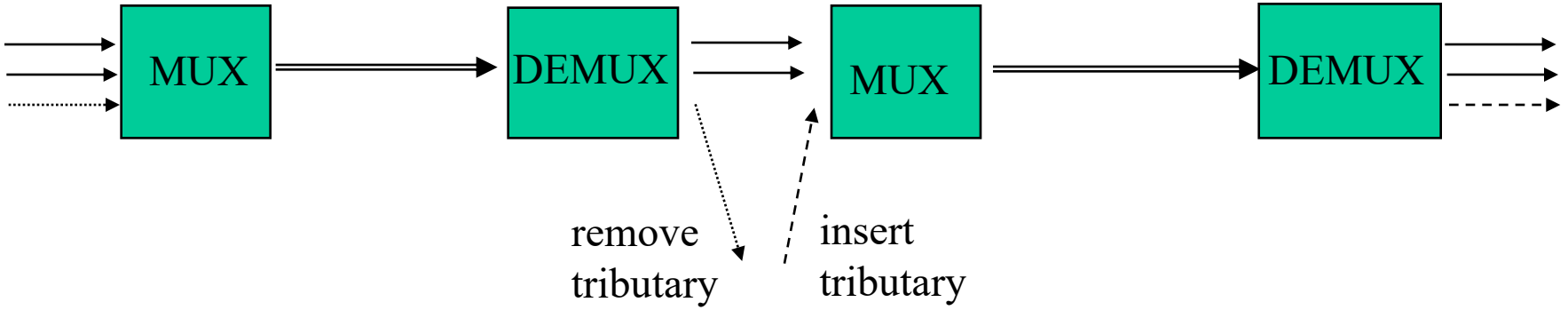
(b) Loop-around in response to fault

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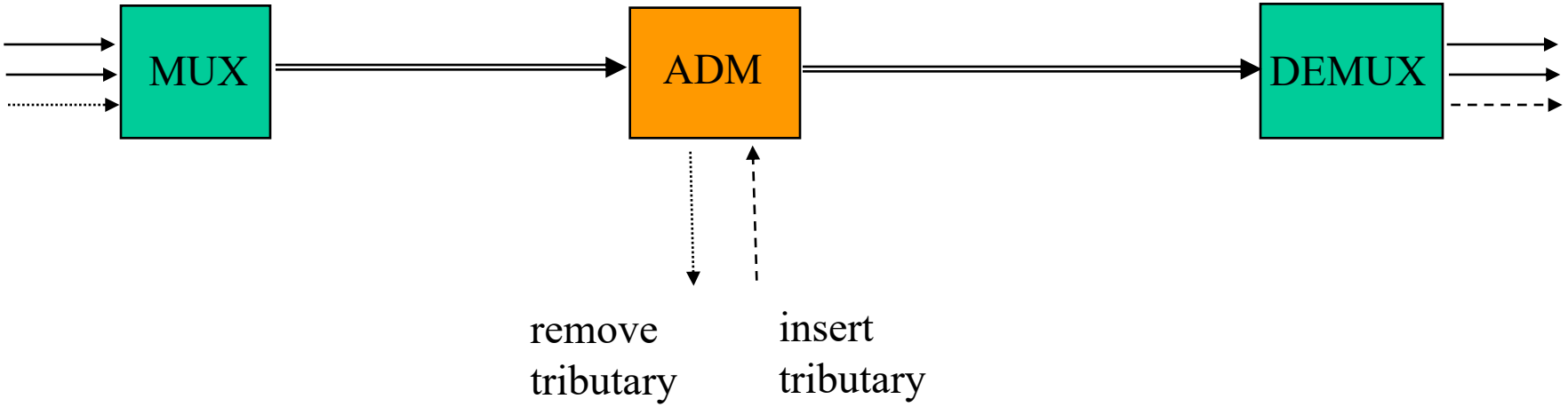

TWO FIBER
BI-DIRECTIONAL
LINE SWITCHING RING
half working
half protection
ADM
half working
half protection
ADM
ADM
ADM

SONET Ring

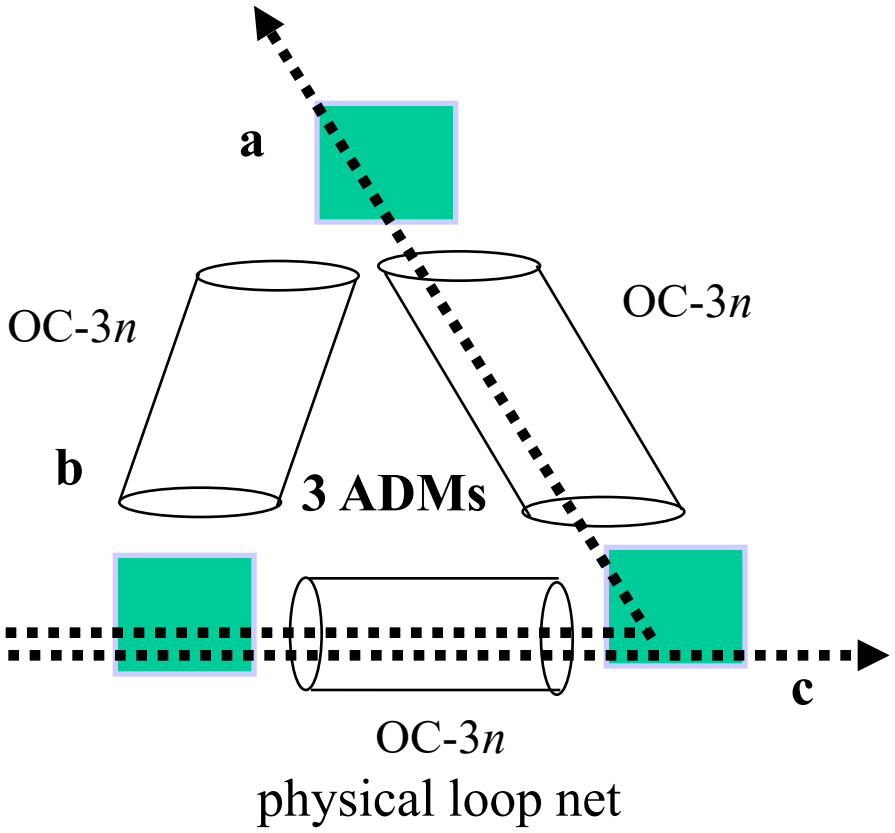
(a) pre-SONET multiplexing



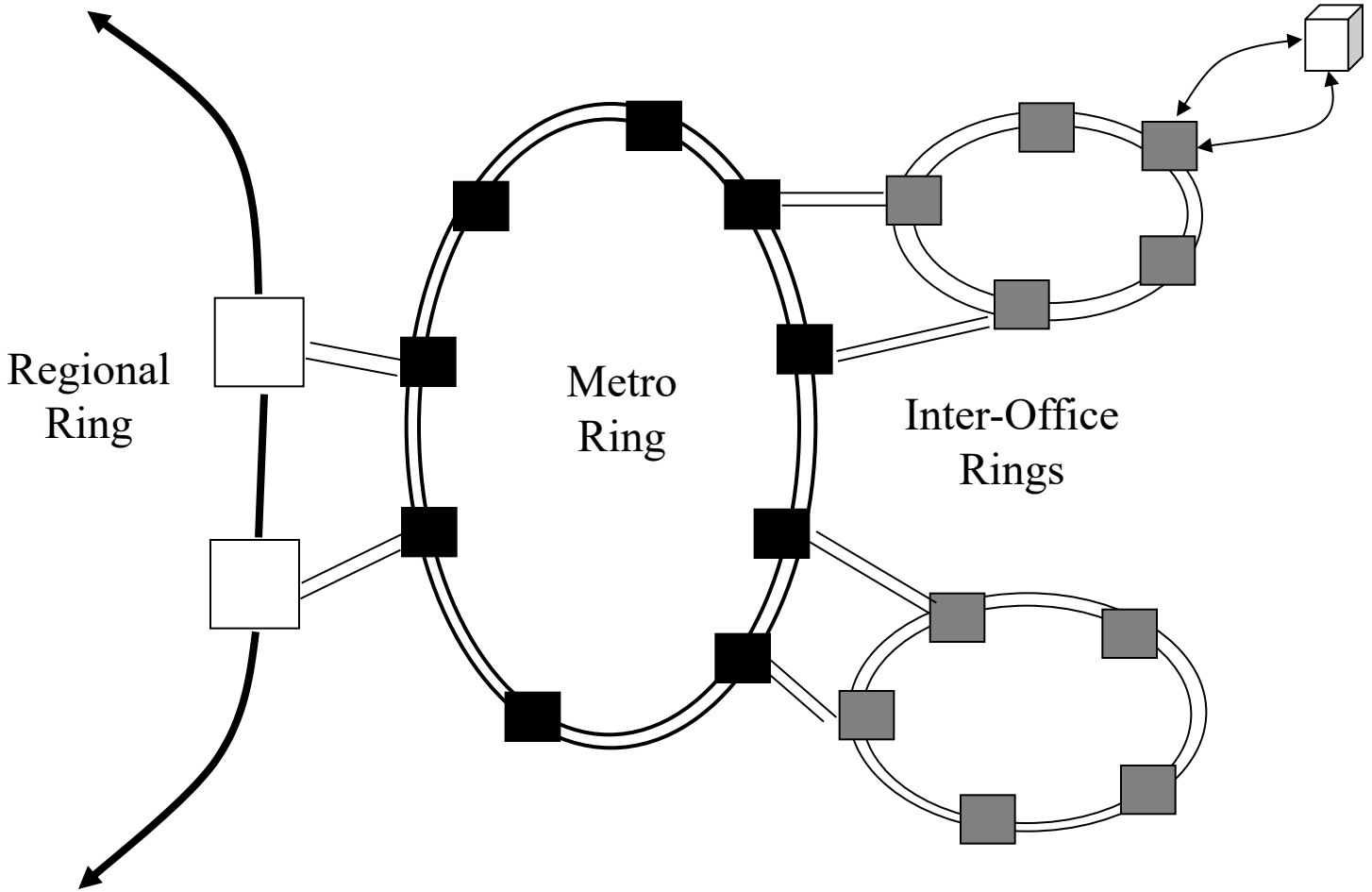
(b) SONET Add-Drop multiplexing



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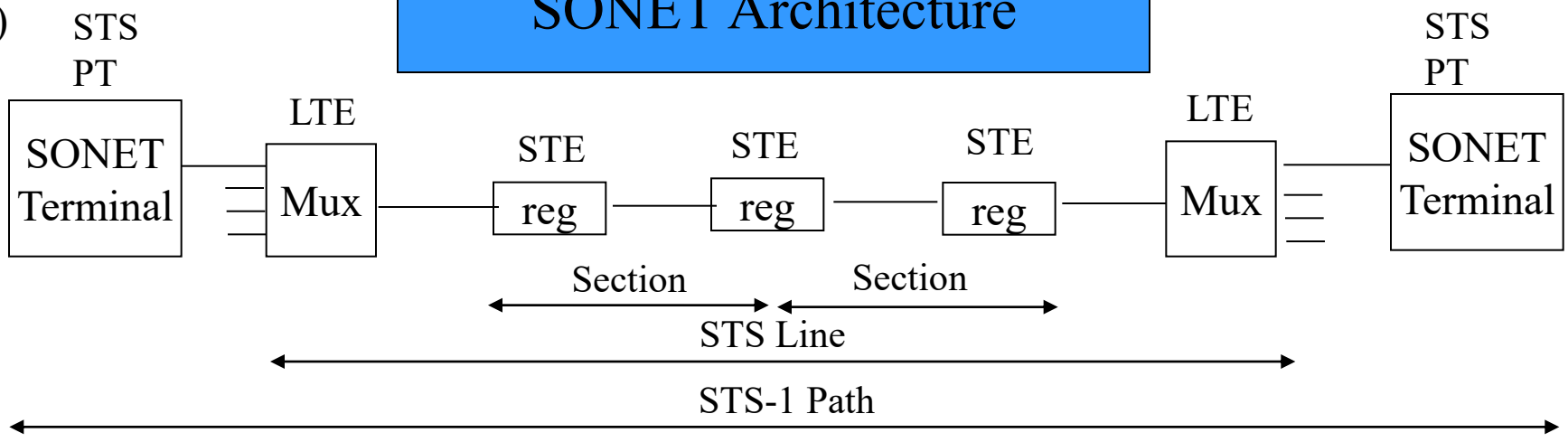


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Figure 4.13

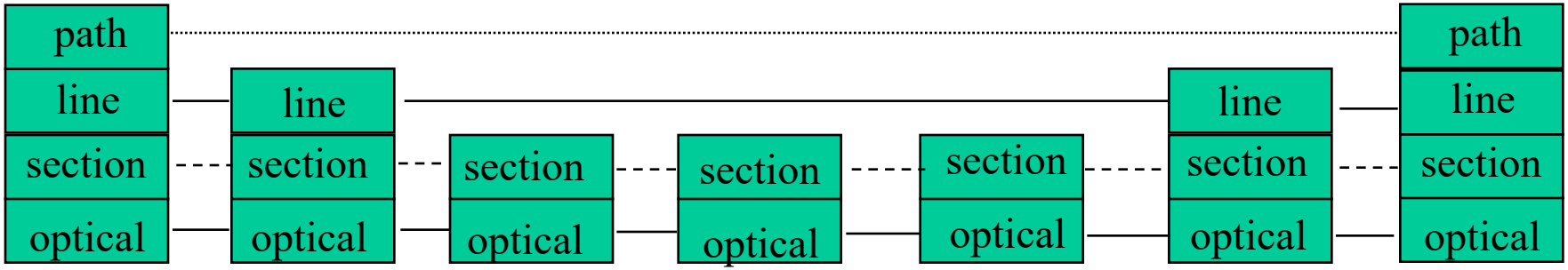
SONET Architecture

(a)

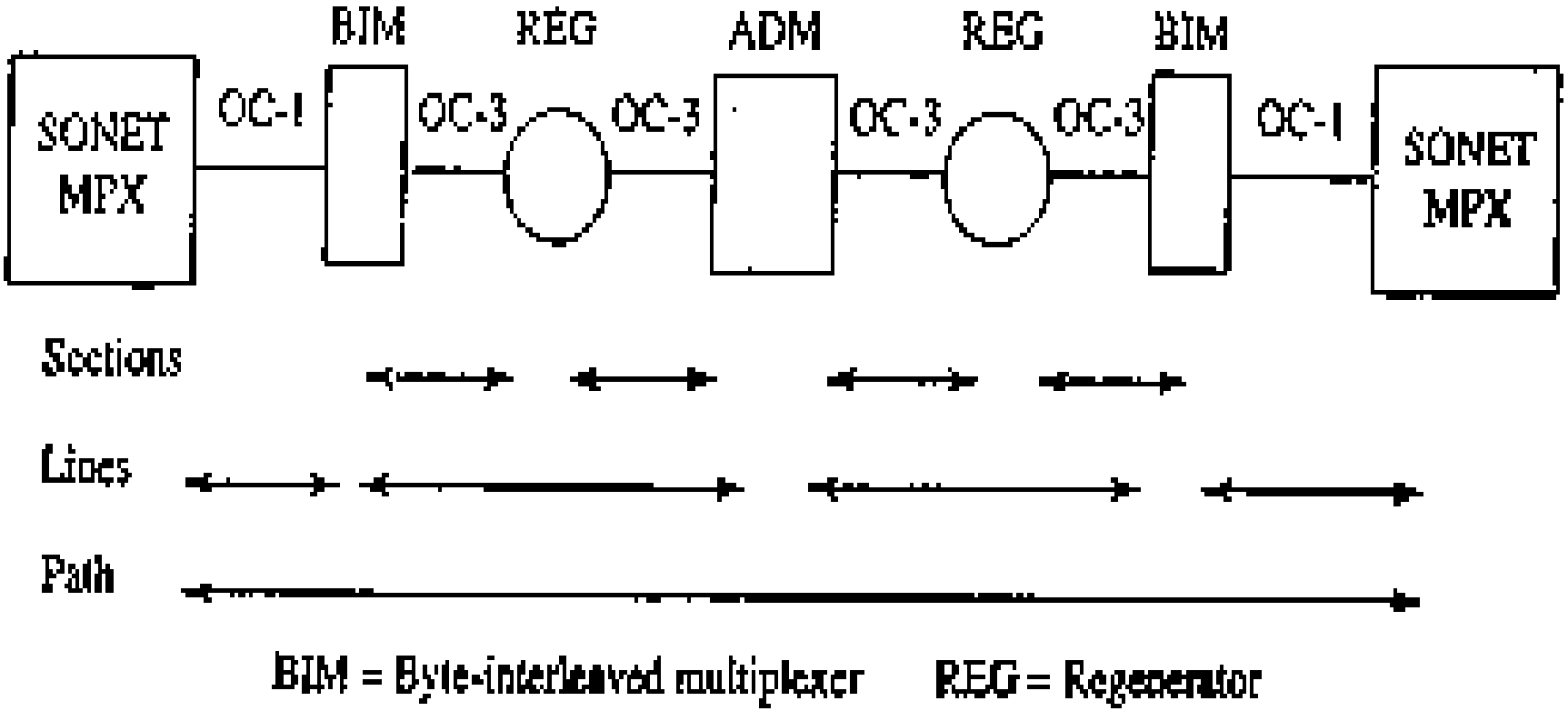


STE: Section Terminating Equipment, e.g. a repeater
LTE: Line Terminating Equipment, e.g. a STS-1 to STS-3 multiplexer
PTE: Path Terminating Equipment, e.g. an STS-1 multiplexer

(b)



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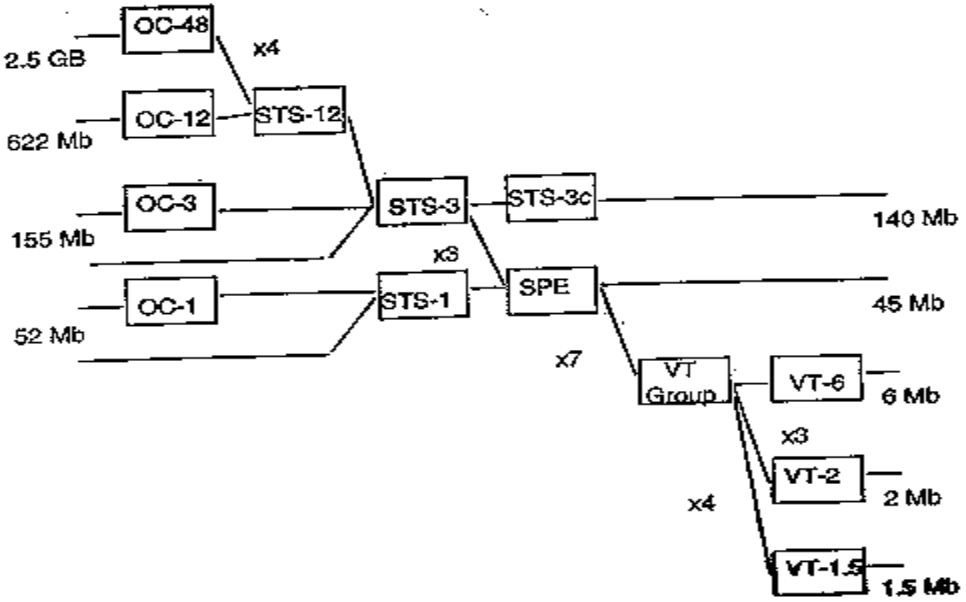
The main SONET network elements.

Mux = BIM (Byte Interleaved Multiplexer)

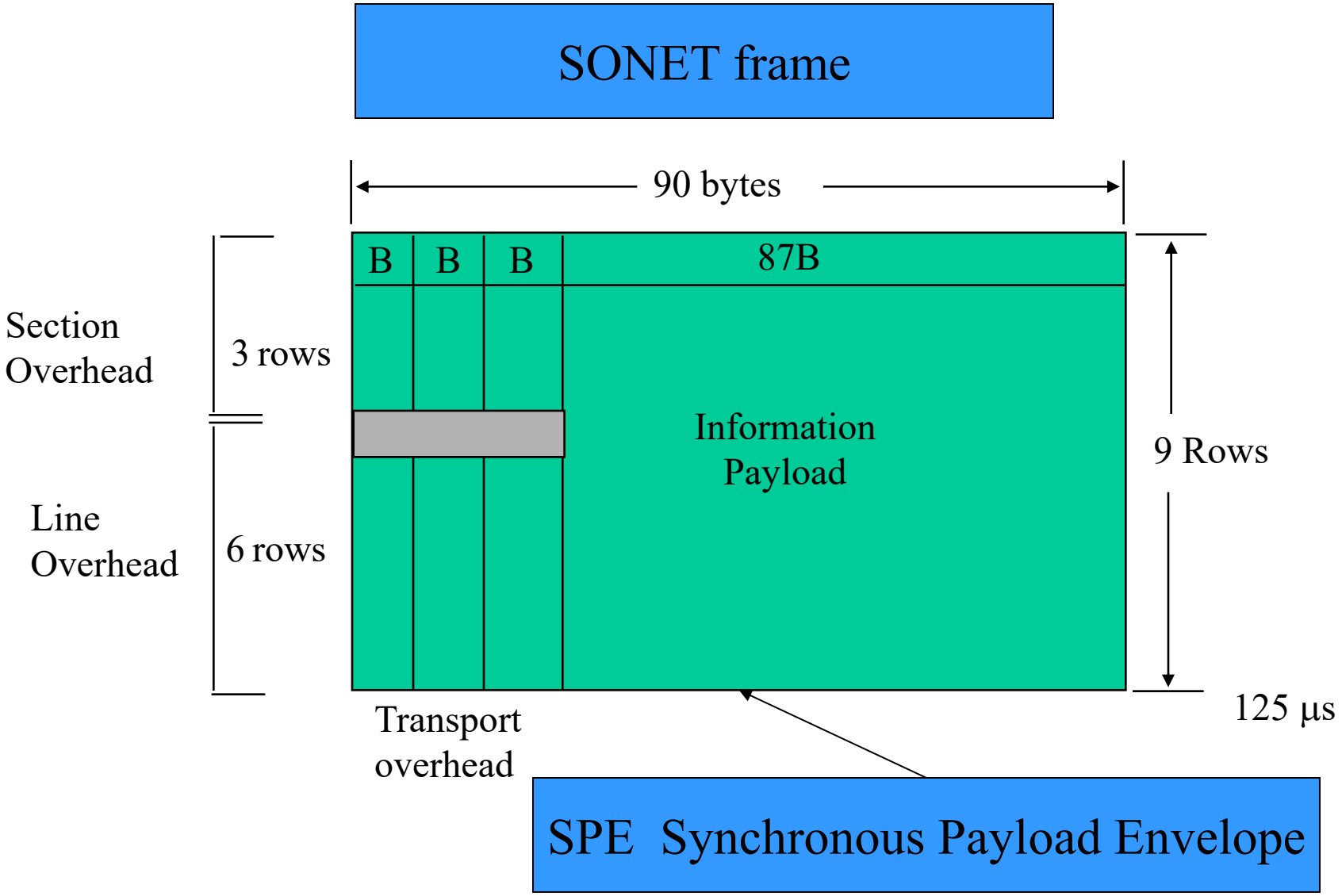
Reg = Regenerator

- Boosters power of optical signal
  - Optical signal *converted to* electrical signal
  - Amplify electrical signal
  - Amplifed electrical signal *converted back to* optical signal

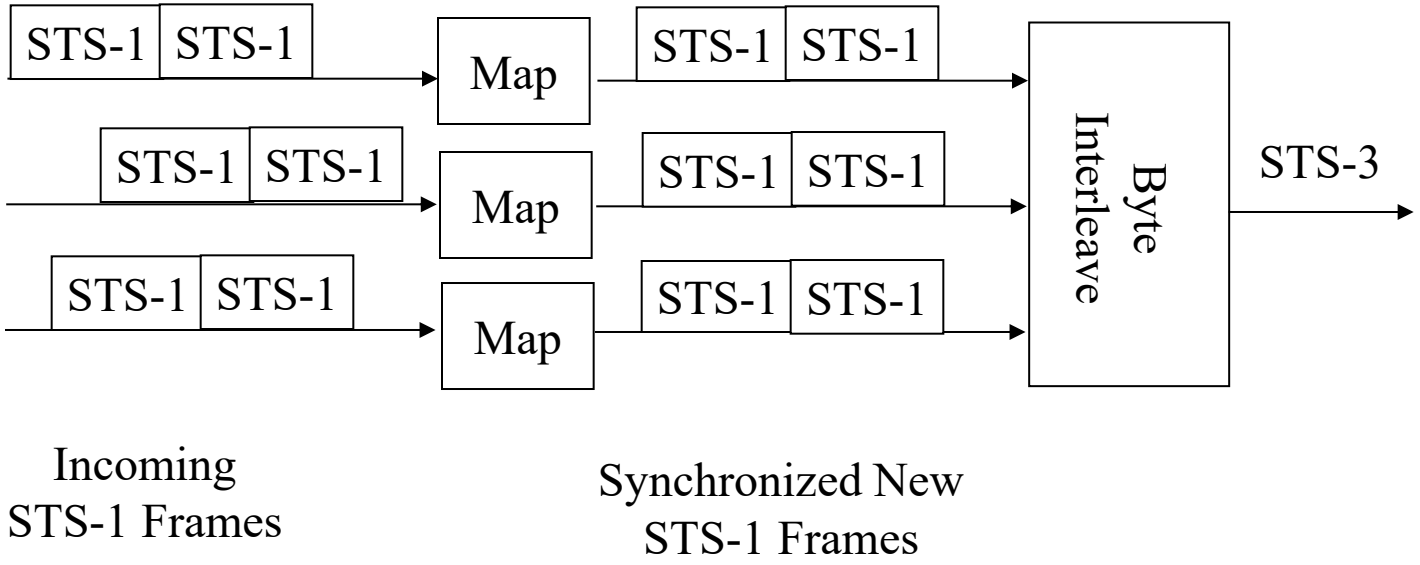
MULTIPLEXING HIERARCHY



- *SERVICE ADAPTORS* MAP VARIOUS SERVICES (VOICE, DATA, VIDEO...) INTO THE PAYLOAD ENVELOPE OF VIRTUAL TRIBUTARIES OR STS-1.
- STS-1 SIGNALS ARE MULTIPLEXED INTO STS-N (BYTE INTERLEAVE SYNCHRONOUS MULTIPLEXER)



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